Version (release date) Coupled Model version	CAM4 (April 2010) CCSM4	CAM5 (June 2011) CESM1	CAM6 (June 2017) CESM2	CAM7 CESM3
PBL Scheme	HB	UW	CLUBB	CLUBB+ (progn. momentum fluxes)
Shallow convection	Hack	UW	CLUBB	CLUBB+ (progn. momentum fluxes)
Deep convection Scheme	ZM_mod1	ZM_mod1	ZM_mod2	ZM_mod3 | CLUBB-MF (ultra high res; ~3km)
Microphysics Scheme	RK	MG1	MG2	PUMAS
Macrophysics Scheme	Zhang	Park	CLUBB	CLUBB+ (prognostic momentum fluxes)
Radiation Model	CAMRT	RRTMG	RRTMG	RRTMG**P**
Drag/Gravity wave source	McFarlane orographic GW scheme	NOGW/TMS oro	Beljaars/NOGW/AnisoOGW	Beljaars/NOGW(+moving mountains source)/ AnisoOGW
Chemistry Package	–	MOZART	MOZART	MOZART
Aerosols Model	BAM	MAM3/CI	MAM4/CI	MAM4 (if using strat. chem. then MAM5)/CI
Dynamical coress	**FV** | EUL | SLD	**FV** | EUL | SLD | SE	**FV** | EUL | SE | FV3	**SE-CSLAM** | MPAS | (SE-NH) | FV
Model top / #levels	~42km /26	~42km / 30	~42km / 32	~80km/93 (Mid Top: **MT**) & ~42km/58 (Low Top: **LT**)
Land/Ocean Model	CLM4/POP2	CLM4/POP2	CLM5/POP2	CLM6/MOM6

HB - Holtslag-Boville **TMS** – Turbulent Mtn Stress
NOGW – non-orographic gravity wave (**GW**); Sources: frontal and convective
{Iso/Aniso}OGW – isotropic/anisotropic orog. GW
CI - cloud droplet activation and ice nucleation
FV/FV3 - finite-volume dycore lat-lon/cubed-sphere

MG – Morrison Gettelman **UW** – U.Washington
ZM - Zhang-McFarlane
CLUBB - Cloud Layers Unified By Binormals
CLUBB-MF - Unified deep-shallow convection
SE - spectral-element dynamical core on GLL grid
SE-CSLAM - SE with CSLAM and physics grid

RT - Radiative Transfer
RRTMGP – Rapid Radiative Transfer Model
MAM – Model Aerosol Model
RK – Rasch-Kristjánsson
EUL/SLD - spectral-transform dycore
MPAS - Model for Prediction Across Scales